

**Faculty of Natural and Applied Science,
Department of Software Engineering,
Veritas university, Bwari-Abuja.**

Course: Digital logic design

(IFT 211)

Digital logic design

- Digital logic design is the process of creating digital circuits and systems using logical operations and binary data (0s and 1s). It involves designing and implementing digital electronics, such as computers, smartphones, and other digital devices, using logic gates, flip-flops, and other digital components.
- Logic design refers to the process of creating and implementing logical systems, circuits, or digital systems that use logical operations to process information.
- Digital logic design is a fundamental aspect of computer science, electrical engineering, and electronics engineering.

- **Digital System:** A digital system is a type of electronic system that processes and represents information using discrete/separate values or digits (0s and 1s) in binary form. It operates on digital signals, which are signals that have distinct values.
- Digital systems are used in modern electronic devices to perform a wide range of tasks, from simple calculations to complex computations and data processing.
- **Some Examples are:**
 - 1. Computers 2. Smartphones 3. Digital Watches 4. Calculators 5. Gaming Consoles

- **Analogue System:** An analogue system is a type of electronic system that processes and represents information using continuous signals.
- Analogue systems are used in applications such as audio equipment, medical devices, and control systems, where signals need to be represented in a continuous manner instead of digits.
- **Some examples are:**
 - 1. Motorcycle speedometer, 2. Analogue Clocks, 3. Thermometers (mercury-based), 4. Rain gauge, 5. Pressure meter

Differences between Digital and Analogue Systems

- 1. Digital systems use discrete signals (0s and 1s), while analogue systems use continuous signals.
- 2. Digital systems represent data in binary form, while analogue systems represent data in a continuous waveform.
- 3. Digital systems are generally more accurate and less prone to errors, while analogue systems can be affected by noise and interference.

- 4. Digital systems are more flexible and can be easily reprogrammed, while analogue systems are often designed for a specific purpose.
- 5. Digital systems can store large amounts of data in a compact form, while analogue systems may require more physical space to store information.
- 6. Digital systems are more resistant to noise and interference, while analogue systems can be more susceptible to signal degradation/distortion.



Digital and analogue thermometers



Digital and analogue clocks

- **Applications of digital logic design**

- Some Applications of digital logic design are:
- 1. Computers: Digital logic design is used to design and develop the central processing unit (CPU) and other digital components of computers.
- 2. Mobile Devices: Digital logic design is used in the development of mobile devices, including processors, memory, and interface controllers.
- 3. Embedded Systems: Digital logic design is used in embedded systems, such as traffic lights, and industrial control systems
- 4. Digital Signal Processing: Digital logic design is used in digital signal processing applications, such as audio and image processing.

- 5. Networking and Communication: Digital logic design is used in network routers, switches, and other communication devices.
- 6. Medical Devices: Digital logic design is used in medical devices, such as patient monitors, and portable medical devices.
- 7. Industrial Automation: Digital logic design is used in industrial automation such as supervisory control and data acquisition systems.
- 8. Aerospace and Defense: Digital logic design is used in aerospace and defense applications, including navigation systems, communication systems, and missile guidance systems.

Information representation and number system

- **Information representation**
- Information representation is the process of encoding data in a format that can be understood and processed by computers. Information representation deals with the the way data is encoded, stored, and communicated.
- In digital systems, it involves converting analog signals into binary code, which is a sequence of 0s and 1s.
- Encoding is the process of converting an information or data into codes or encrypted form.

Importance of data encoding

- It helps to compress the data for easy transfer or storage
- It helps to maintain or protect the data integrity
- It ensures the confidentiality of data
- To protect data or information from unauthorized access or use.
- To ensure compliance with data sharing policy or regulations.

Types of Information Representation

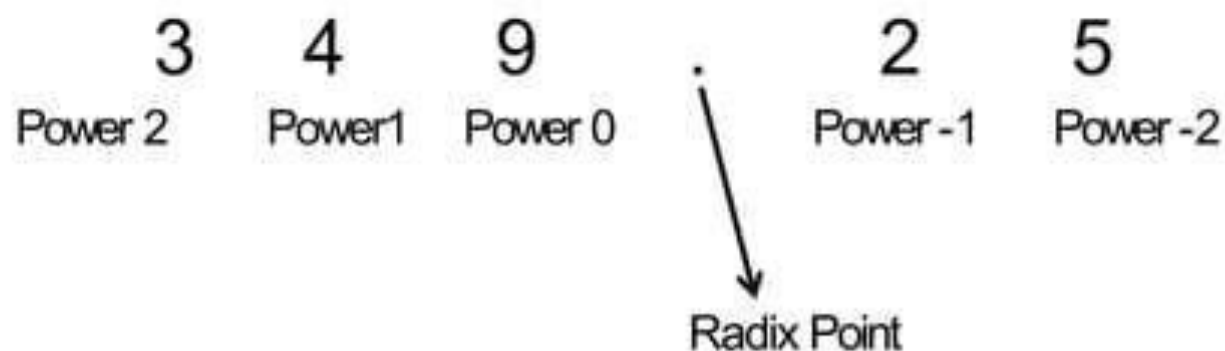
- **Image Representation:** Image data is represented using raster graphics (bitmap) or vector graphics.
- **Audio Representation:** Audio data is represented using digital audio formats such as WAV, MP3, and AAC.
- **Video Representation:** Video data is represented using digital video formats such as MPEG, AVI, and MP4.
- **Text Representation:** Text data is represented using character encoding schemes such as ASCII (American Standard Code for Information Interchange: 7-bit encoding scheme), Unicode, and UTF-8.
- **Numerical Representation:** Numerical data is represented using number systems such as binary number.

Introduction to Number System

- ❑ Number system is basis for counting various items.
- ❑ Modern computer operated in binary number system.
- ❑ Decimal, Octal, Hexa, BCD represents more bits in a binary numbers in a compressed form.
- ❑ Base/Radix of the number systems are 2, 8, 10, 16
- ❑ Base decides the total no of digits available in a system
 - E.g Binary Base 2 0 or 1

Introduction to Number System

- Each digit position represent different multiples of base.



Different Number Systems



□ Decimal	Base 10	0 to 9
□ Binary	Base 2	0 or 1
□ Octal	Base 8	0 to 7
□ HexaDecimal	Base 16	0 to 15

Decimal Number System

- In decimal number system we can express any decimal number in units, ten, hundred and thousand.
- E.g Represent decimal number 98.72 in power of 10

$$\begin{array}{ccccccc} & 9 & & 8 & & . & 7 & & 2 \\ N = & 9 \times 10^1 & + & 8 \times 10^0 & + & 7 \times 10^{-1} & + & 2 \times 10^{-2} \end{array}$$

Binary Number System

- E.g Represent binary number 1101.101 in power of 2 & find its decimal equivalent.

$$\begin{aligned} \square N &= 1 \quad 1 \quad 0 \quad 1. \quad 1 \quad 0 \quad 1 \\ &\quad 1*2^3 + 1*2^2 + 0*2^1 + \dots\dots\dots \\ &\quad 8 + \quad 4 + \quad 0 + \quad 1 + \quad 0.5 + \quad 0 + \quad 1.25 \\ N &= (13.625)_{10} \end{aligned}$$

Octal Number System

- Represent 0-7 digits in 3 bit format 000-111
- E.g Represent octal number 567 in power of 8 & find its decimal equivalent.

$$\begin{aligned} N &= \quad \quad 5 \quad \quad 6 \quad \quad 7 \\ &5 \cdot 8^2 + 6 \cdot 8^1 + 7 \cdot 8^0 \\ &320 + 48 + 7 \\ &(375)_{10} \end{aligned}$$

Hexadecimal Number System

Invalid state for Decimal

DECIMAL	BINARY	HEX
0	0000	0
1	0001	1
2	0010	2
3	0011	3
4	0100	4
5	0101	5
6	0110	6
7	0111	7
8	1000	8
9	1001	9
10	1010	A
11	1011	B
12	1100	C
13	1101	D
14	1110	E
15	1111	F

Hexadecimal Number System

- 4 bits representation 0000 - 8421
- Represent hexadecimal number 3FD in power of 16 & find the decimal equivalent.

$$\begin{aligned} \square N &= 3 \qquad \qquad F \qquad \qquad D \\ & \quad 3 \cdot 16^2 + 15 \cdot 16^1 + 13 \cdot 16^0 \\ & \quad 768 \quad + \quad 240 \quad + \quad 13 \\ & \quad (1021)_{10} \end{aligned}$$

Binary to Octal Conversion

□ Convert 111101100 to octal

□ **Solution:**

111	101	100
7	5	4

$(754)_8$

Octal to Binary Conversion

□ Convert $(634)_8$ to Binary

□ Solution:

6	3	4
110	011	100

$(110011100)_2$

- Solution:**
- | | | | |
|------|------|------|------|
| 1101 | 1000 | 1001 | 1011 |
| D | 8 | 9 | B |

(D89B)₁₆

Hex to Binary

□ Convert $(3FD)_{16}$ to Binary

□ Solution: 3 F D
 0011 1111 1101

$(001111111101)_2$

Octal to Hexadecimal

□ Convert (615)₈ to hex

□ Solution : 6 1 5
 110 001 101 Binary

Now we will change binary into hex

3 zero's added to
the left ← 0001 1000 0101
 1 8 B

Hexa to Octal

□ Convert $(25B)_{16}$ to octal

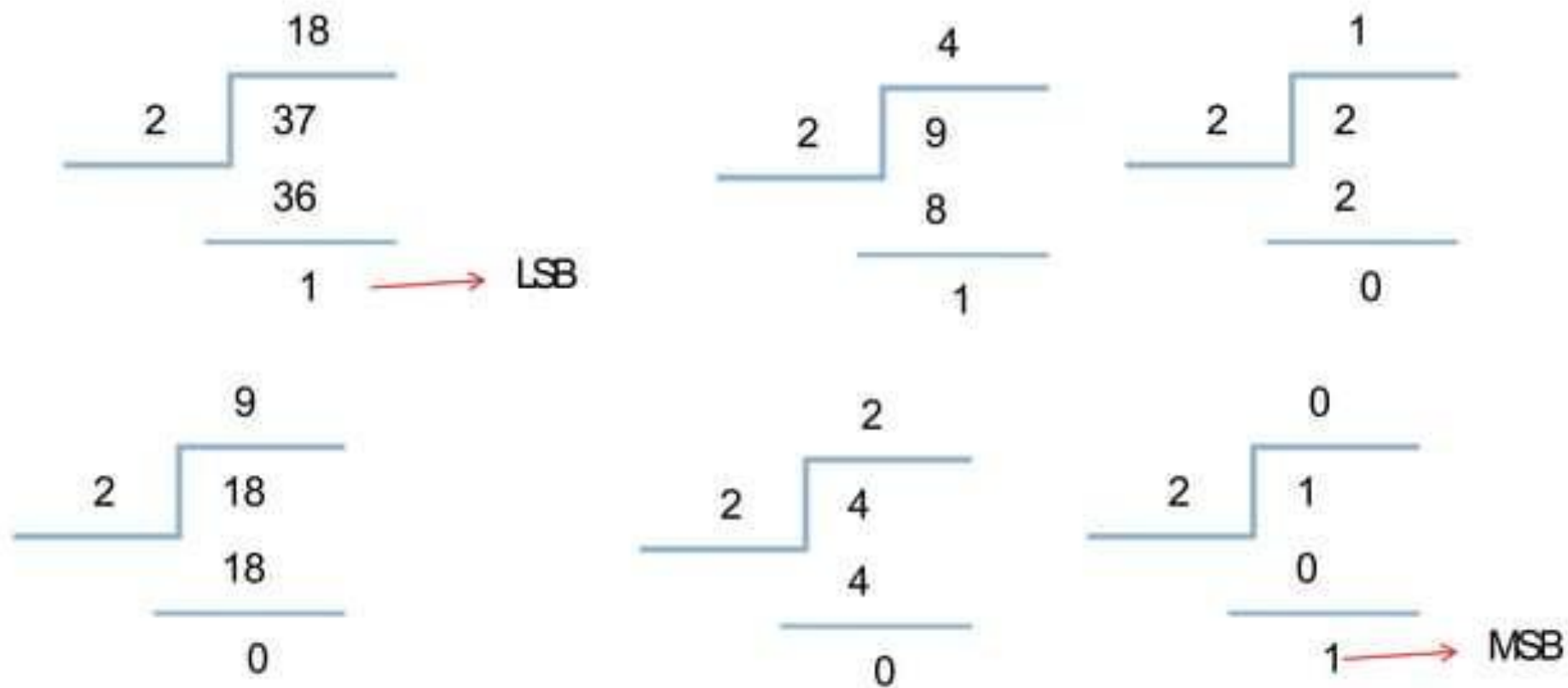
Solution : 2 5 B
 0010 0101 1011 Binary

Now we will convert into octal (Set of three bits)

 001 001 011 011
 (1 1 3 3)₈

Example

□ Convert decimal 37 into binary



Binary = 100101

Self Task

- Convert decimal 214 to octal
- Determine Base 1st
- Ans = $(326)_8$

Class Exercises

- Convert $(475.25)_8$ to decimal
- Convert $(9B2.1A)_{16}$ to decimal
- Convert $(3102.12)_4$ to decimal